

Where Does Physics Help a Graph Network?

A Leakage-Controlled Study of SEIR-Informed Spatio-Temporal GNNs for Dengue Forecasting

Tharusha Perera

Praveen De Silva

Janith Mahanama

Bimsara Udurawana

Maleesha Kumarasinghe

Praveen Nawarathna

Department of Computer Science and Engineering, University of Moratuwa

Moratuwa, Sri Lanka

{tharushap.23,praveend.23,janithm.23,bimsarau.23,kumarasinghemp.23,praveenn.23}@cse.mrt.ac.lk

ABSTRACT

Spatio-temporal graph neural networks (ST-GNNs) and SEIR-informed neural models are both proposed for district-level dengue forecasting, but they are rarely compared under one leak-free protocol. We study whether replacing the LSTM of an SEIR-informed forecaster with a graph encoder helps, on Sri Lanka’s weekly record (25 districts, three-week horizon). Before modelling, we audit the benchmark array used by prior work. Its rows are out of date order, every training split contains 2023 data from after the test periods, its case and climate columns drift apart by up to a year, its “lagged” climate channels hold values 12–17 weeks in the *future*, and its largest spike is a spreadsheet error. We rebuild a 559-week dataset from the source reports and enforce causal lags with tests. On one harness, six encoders (five published ST-GNNs and the SEIR-LSTM’s LSTM) are each paired with a direct head, a pure SEIR decoder and a gated SEIR head. The architecture matters little: working encoders span 0.15 RMSE, and the graph adds 0.07. The pure SEIR decoder is worse than persistence on every encoder, and 6–7 RMSE worse than the direct head on each working one. The cause is structural: 14–16% of targets lie below its zero-transmission floor, and its force of infection is only $r^2 \approx 0.25$ predictable. A gated SEIR head, which lets the simulator supply only the departure from persistence, brings all six encoders below persistence and repairs the three that fail as published by 5.7–17.2 validation RMSE on 9/9 paired runs. A pre-registered control shows that this repair is not physics. The same head with the simulator removed accounts for 96–112% of it. With the simulator added, validation RMSE is significantly worse on one failing encoder and not better on the other two, and 0.74–0.88 worse on the working ones. A negative-binomial likelihood is the largest controlled gain (−0.41, 9/9). The graph beats the LSTM only inside the gated SEIR head (9/9 on three origins), and that advantage does not replicate on nine disjoint origins. More than twenty further levers are not adopted: architecture, embeddings, climate lags, synthetic data, a spatial physics penalty and a metapopulation SEIR head. Their residuals correlate 0.87–0.99. On this target the useful prior is persistence, not transmission dynamics. Anchored models, SEIR-informed ones included, do score better out of sample, but validation does not select them.

KEYWORDS

dengue forecasting, spatio-temporal graph neural networks, SEIR, physics-informed learning, data leakage, reproducibility, negative results

1 INTRODUCTION

Dengue is the fastest-spreading mosquito-borne viral disease, and Sri Lanka faces recurring severe epidemics: the 2017 outbreak alone recorded over 186,000 cases and 440 deaths [15]. Roughly three weeks separate infection from diagnosis. District-level forecasts issued $H = 3$ weeks ahead therefore carry actionable lead time for clinical capacity and vector control.

Two families of model are proposed for this task. Spatio-temporal graph neural networks, transferred from traffic forecasting, treat districts as nodes and learn from neighbours; Weng et al. [17] benchmark five of them on Sri Lanka’s record. SEIR-informed models push the forecast through compartmental dynamics; Liu et al. [7] drive an SEIR model with an LSTM on the same country. The natural hybrid is an *SEIR-GNN*: the SEIR-LSTM with its LSTM replaced by a graph encoder, so that spatial coupling and mechanism act together. Whether that hybrid helps, and through which component, is the question of this paper.

One property of the data governs every answer: weekly incidence is strongly autocorrelated. On the corrected data, cases at $t - 1$ explain $r^2 = 0.89$ of cases at t , while the best single climate covariate at a causal lag explains 0.03. Every model competes with the naive persistence forecast $\hat{y}_{t+h} = y_t$. A benchmark that omits it cannot tell whether a model contributed anything, and the reference benchmark omits it.

Our first SEIR-GNN pipeline reported a win, and it was wrong. Its headline set the test error at one forecast origin (26.1) against persistence averaged over three (36.0); at that same origin persistence scored 22.4, and it won at every origin. The pipeline also trained for about sixty gradient steps, optimised SMAPE while scoring RMSE, held the force of infection constant across the horizon, and collapsed every covariate to a scalar before the graph saw it. This paper reports the controlled re-run. Its protocol and each experiment’s decision rule were committed before the runs, and negative results are reported as such.

Contributions.

- **A data audit and a corrected dataset** (§3). The benchmark array mixes time orders, puts post-test data in every training split, and feeds future climate as “lagged” features. We rebuild 559 weeks from source reports, with causal lags enforced by perturbation tests.
- **One harness for six encoders and four SEIR-informed heads** (§4), with validation-only selection, paired tests at two units, and a nine-origin confirmation.
- **Whether physics helps** (§5). A pure SEIR decoder fails on every encoder, for reasons we measure. A gated SEIR head appears to repair the three encoders that fail as published, but a pre-registered control attributes 96–112% of that repair to the persistence anchor and none of it to the simulator (§5.2).
- **An informational ceiling, measured** (§5.6). More than twenty architectural, data and physics levers leave the same errors behind (residual correlation 0.87–0.99).

2 RELATED WORK

ST-GNNs for dengue. Spatio-temporal graph networks couple a graph operator over regions with a temporal operator over history. The five we evaluate come from traffic forecasting. STGAT and A3TGCN [1] use attention, ASTGCN [4] attends over space and time, AAGCN [14] adapts its adjacency, and DCRNN [6] diffuses over the graph inside a recurrent cell. Weng et al. [17] benchmark all five on Sri Lanka’s dengue record, and GulMohamed et al. [3] add per-horizon heads, temporal attention and a Gaussian output.

Mechanistic hybrids. Liu et al. [7] forecast Sri Lankan dengue with an SEIR model whose force of infection is predicted by an LSTM; we call this SEIR-LSTM. For epidemics more generally, EINNs [11] couple a neural forecaster with a mechanistic model through shared latent dynamics. CausalGNN [16] and MepoGNN [2] embed a metapopulation SEIR in a graph network, so that districts are coupled inside the dynamics. Physics-informed networks [10] add dynamics as a loss; MP-PINN [8] applies this to epidemic curves in a regime where data are scarce relative to the dynamics. We use the SEIR–SEI stage durations of Phaijoo and Gurung [9].

Baselines and remedies from forecasting. Simple baselines often match deep spatio-temporal models: STID [13] replaces graph operators with identity embeddings. Reversible instance normalisation [5] targets distribution shift, and count likelihoods such as the negative binomial of DeepAR [12] target skewed demand. TimeGAN [18] generates synthetic sequences for data-poor settings. We test each of these here, because each addresses a weakness our diagnosis found.

3 DATA

3.1 Reproducing the benchmark

We ran the reference implementation [17] unmodified—their code, their array, their protocol—on a pinned software stack. It recovers their published numbers to within 7.3% across twenty comparisons, with AAGCN and DCRNN matching to four decimal places. The published results are real, but three properties of the evaluation account for the reported advantage. The column labelled “cross

Table 1: Audit of the benchmark array. Each finding is confirmed against the source reports or an independent reanalysis.

finding	consequence
Rows 0–47 are 2023; the rest run 2013–2022	every training split of the standard protocol contains data from <i>after</i> its test period
Case column reordered, climate not	cases sit beside weather from a different week, drifting to 48–55 weeks apart by 2019
“Lagged” precipitation and canopy channels	match ERA5 best 12–13 weeks <i>later</i> ; min-NDVI best 9–18 weeks later: future information
49 source reports missing, one week duplicated	2014 half missing; 2022 stops in March
Week-395 “19× spike”	a formula dragged across one printed table (15 of 24 cells are each the sum of the two before); the same table’s cumulative row gives the true count
4% of weather values are zero-filled	Jaffna’s columns are all zero: it gets no grid cells in the authors’ aggregation

validated” is computed on a loader spanning the full series, so it includes training data. RMSE is averaged per window rather than pooled, which on this target is ~39% lower. No naive baseline is reported, and adding persistence to that same column reverses the ranking: it beats *every* model on MAE and RMSE. An independent re-implementation that shares no code with either codebase reproduces this.

3.2 Auditing the benchmark array

The benchmark array (`sri_lanka_2013–2022_shifted.npy`) has no date axis. We dated every row against the 552 weekly epidemiological reports it was built from, and dated the climate channels against ERA5 reanalysis and the authors’ own raw satellite files. Table 1 lists what we found.

Two of these are information leaks. Training on 2023 while testing on 2018–2022 lets a model see the future of its test period. “Lagged” climate channels that carry rainfall and vegetation from three to four months ahead give it that future’s weather. Any score on this array, including the published ones and our own earlier work, is therefore an optimistic bound. Earlier drafts of this study called the week-395 spike an administrative reporting backlog. It is a spreadsheet error, and it can be corrected from the same published table.

3.3 The corrected dataset

We rebuilt the case series from every source report, keyed by report volume and number: 559 weeks from 2013-W26 to 2024-W10. The seven weeks with no report stay missing and are never interpolated. All 451 weeks the array shares with the rebuilt series match exactly, so the array’s values were correct and only their order and coverage were wrong. Covariates are aligned row for row: ERA5 temperature, precipitation, humidity and soil moisture for each district; MODIS NDVI taken *as of* each week, so a composite counts only after its release; and population using the previous year’s official figure.

Lags are applied in the model, not baked into the data: cases up to $t - 1$, climate up to $t - 2$, and NDVI and population as of t . A test perturbs every input value at or after each limit and asserts that no model input moves, and every new input path in this study received its own such test. The 2022–23 serosurvey and serotype-switch dates are used for validation only. Unless stated otherwise, every number from here on is on this dataset. The persistence test RMSE on the frozen three origins (averaged over origins, as for every model) moves from 44.80 (29.52 with the week-395 windows excluded) on the array to **36.02**, and it reproduces exactly when recomputed from the raw series. The last test fold now covers the 2023 epidemic.

4 MODELS AND PROTOCOL

A run is a tuple (encoder, head, likelihood, features). Each element is varied while the others are held fixed, so every difference is attributable to one component.

4.1 Encoders

The five published ST-GNNs (STGAT, A3TGCN, ASTGCN, AAGCN, DCRNN) are built from the reference implementations. The sixth encoder is the two-layer LSTM of the SEIR-LSTM [7]. Each maps a $W = 3$ week window of log cases, plus any exogenous features, to a 64-dimensional representation h_i per district. The graph is the benchmark’s district adjacency with self-loops, row-normalised; AAGCN runs with its adaptive adjacency off, as published.

4.2 Heads

All heads output fold-standardised $\log(1 + y)$ for $h = 1, 2, 3$. Let p_i be the persistence anchor (last observed week) in the same space and $f(h_i)$ a linear read-out.

- **direct**: $\hat{z}_i = f(h_i)$, as in the published ST-GNNs.
- **residual**: $\hat{z}_i = p_i + f(h_i)$.
- **gated**: $\hat{z}_i = p_i + \sigma(a)(f(h_i) - p_i)$, a single learned gate initialised at $\sigma(-2) = 0.12$. This head was added for §5.2; it is the gated SEIR head with the simulator removed.
- **SEIR decoder** (foi): $f(h_i)$ gives a weekly force of infection $\lambda_{i,h} = \exp(f(h_i) + b)$, clamped at 1/7 per day. An SEIR model is stepped daily from an initial state seeded by observed cases: E_0 from cases at $t - 2$ scaled by the reporting rate $\rho = 1/11$, $S_0 = 1 - 0.682$ from a 2013–14 Colombo serosurvey, incubation rate 0.1/day and recovery rate 1/7/day. Predicted incidence times ρ times population gives counts. A learned scale on E_0 and ρ is allowed.
- **gated SEIR** (foi_res): $\hat{z}_i = p_i + \sigma(a)(\text{SEIR}(\lambda_i) - p_i)$. The simulator supplies the *departure* from persistence. This is our SEIR-GNN when the encoder is a graph network, and SEIR-LSTM when it is the LSTM.
- **metapopulation SEIR** (foi_meta): the encoder predicts a transmission rate $\beta_i(t)$, and the simulator recomputes the force of infection each day as $\beta_i \sum_j C_{ij} I_j / N_j$, with $C = \text{softmax}(\log A + \Theta)$ a learned row-stochastic coupling initialised at the border graph A , as in MepoGNN and CausalGNN [2, 16]. It is gated like foi_res.

4.3 Likelihood, features and physics penalty

The default objective is squared error on the standardised target. The negative-binomial (NB) alternative adds a dispersion head and minimises the NB2 negative log-likelihood with the mean $\mu = \exp(\hat{z}\sigma + m) - 1$. Under squared error on $\log(1 + y)$, exp of the fitted value is a conditional *median*, while RMSE rewards the conditional *mean*; NB’s μ is the mean. Seasonal features are sin and cos of the week of year at one and two cycles per year. The spatial physics penalty adds $\lambda_s \sum_{(i,j) \in E} (\hat{y}_i/s_i - \hat{y}_j/s_j)^2$ with s_i the district’s training-mean cases: relative incidence should not jump across a border.

4.4 Protocol

Rolling origins at 0.55, 0.70 and 0.85 of the series, three seeds each. The 30 windows before each origin are validation, the following 15% are test, and normalisation uses training weeks only. Windows touching a missing week are dropped, never imputed. Training uses Adam (learning rate 3×10^{-3} , weight decay 10^{-4} , batch 32), early stopping on validation with patience 40, and 300–400 epochs. Each run’s RMSE is pooled over its windows on raw counts, overall and per horizon; tables report the mean over runs.

Selection uses validation RMSE only. Test is written to disk and never used to choose. The protocol, and for each later experiment its arms, control, decision rule and a prediction of the outcome, were committed before it ran. An arm is adopted if its mean validation delta against the control is negative, it wins at least 7 of 9 (origin, seed) units, and it does not raise horizon-3 error.

Two pairing units. Differences are paired on matched (origin, seed) units and tested with an exact two-sided sign-flip permutation test, BH-adjusted within each family. The *origin* unit (seeds averaged) is what a claim about the series rests on, but with three origins its p -value cannot fall below 0.25. The (origin, seed) unit reaches 0.004 but mostly measures run-to-run stability. Claims about the series therefore need the **nine-origin** protocol: origins at $0.40 + k/15$ ($k = 0, \dots, 8$), with nine disjoint test spans of 1/15 of the series, where the origin unit reaches 0.004.

Reproducibility. Runs are single-threaded and bit-reproducible within one environment. Grids ran on local CPUs and on Kaggle CPU kernels that clone a pinned commit. The two environments agree on means but not on single runs, so every comparison pairs arms from the same grid in the same environment.

5 RESULTS

All numbers are RMSE in weekly cases on the corrected dataset. Validation decides and test is reported beside it. Unless stated otherwise, pairs are on the (origin, seed) unit of the frozen three origins.

5.1 Six encoders, one harness

Table 2 and Figure 1 give the comparison the SEIR-GNN question needs, with every encoder sharing folds, loss, early stopping and metric.

Architecture matters little, and the graph less. Behind the direct head AAGCN, ASTGCN and the LSTM score 16.76, 16.84 and 16.91: a spread of 0.15, and all three beat persistence (17.86). The LSTM has no message passing at all. A GCN beats the same network without message passing by only 0.07 (4/9, Table 3). STGAT,

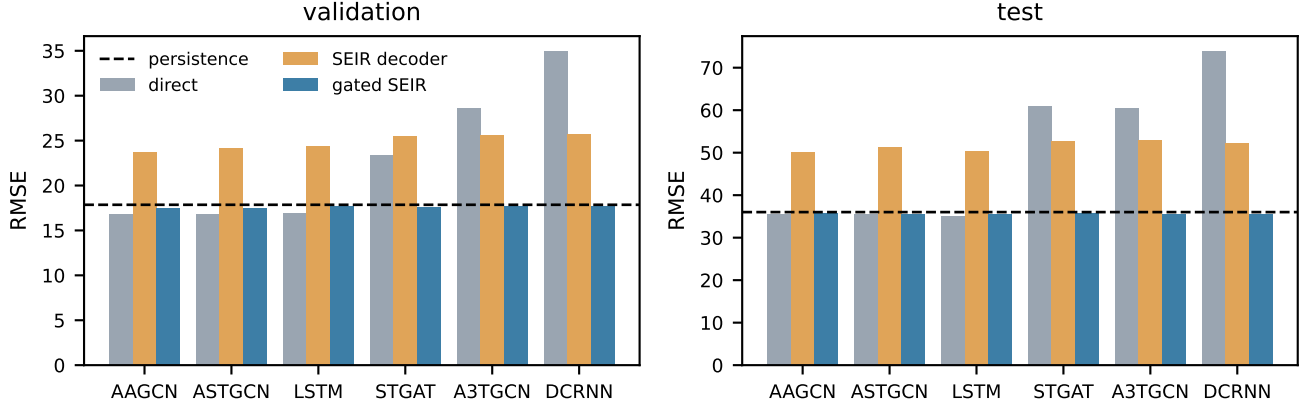


Figure 1: Six encoders behind three heads (Table 2). The direct head separates encoders that work from encoders that fail; the SEIR decoder fails on all of them; the gated SEIR head puts all six at the same level, below persistence (dashed) on validation and test.

Table 2: The six encoders on one harness (corrected data, 3 origins $\times$ 3 seeds, squared error, no seasonal features). Mean validation / test RMSE. The pure SEIR decoder (foi) is worse than persistence on every encoder; the gated SEIR head (foi_res) brings all six to within 0.25 of each other, below persistence.

encoder	direct	SEIR decoder	gated SEIR
AAGCN	16.76 / 35.59	23.69 / 50.06	17.44 / 35.78
ASTGCN	16.84 / 35.39	24.15 / 51.28	17.48 / 35.60
LSTM	16.91 / 35.10	24.32 / 50.25	17.69 / 35.52
STGAT	23.32 / 60.82	25.52 / 52.74	17.62 / 35.84
A3TGCN	28.63 / 60.40	25.55 / 52.92	17.68 / 35.50
DCRNN	34.89 / 73.74	25.72 / 52.28	17.67 / 35.51
persistence	17.86 / 36.02		

A3TGCN and DCRNN score 23.32, 28.63 and 34.89, all worse than persistence; STGAT and A3TGCN are worse at every horizon. A linear model on the same inputs recovers 47–62% of STGAT’s and A3TGCN’s residual variance out of sample but only 4–6% of the working encoders’. The failing encoders are underfitting, limited by their wiring rather than by the data.

The SEIR decoder fails on every encoder. Routing the forecast entirely through the simulator scores 23.69–25.72: 6.9–7.4 worse than the direct head on each working encoder. The deficit is structural, and we measured it without training by inverting the simulator for the force of infection that reproduces each observed count: (i) with transmission switched off, the E already seeded from recent cases still emits *more* cases than observed for 14–16% of targets, which are therefore unreachable at any network output; (ii) the required force of infection is only $r^2 = 0.22$ –0.26 predictable from case history, against 0.81–0.82 for the direct target; (iii) it is not an optimisation problem, since ten times the training budget bought 0.04; (iv) it is not susceptible depletion, since only 2.5% of cells reach the susceptible clamp. Point (ii) replicates on the corrected

data, through a different mechanism, our earlier renewal-equation finding on the benchmark array: the reproduction number is about 26% predictable out of sample.

The gated SEIR head repairs the failing encoders. Letting the simulator supply only the departure from persistence brings all six encoders to 17.44–17.69, within 0.25 of each other and below persistence on validation (17.86) and on test (35.50–35.84 against 36.02). For the three failing encoders this is a large repair, on 9/9 paired units each: STGAT -5.70 (test -24.99), A3TGCN -10.95 (-24.90) and DCRNN -17.22 (-38.23). It costs the working encoders 0.63–0.78 on validation. Read at face value, this is the physics-informed result the study set out to find. The next section shows that it is not.

5.2 Anchor, gate or physics?

The gated SEIR head differs from the direct head in three ways at once: it forecasts a departure from last week’s value (the anchor), it shrinks that departure through a learned gate, and the departure comes from the SEIR simulator. Only the third is physics. A pre-registered control separates them by rerunning all six encoders behind four heads that add these one at a time: direct, residual (anchor), gated (anchor and gate) and gated SEIR (all three). The rescue is credited to physics only if gated SEIR beats its no-physics twin on at least 7 of 9 units with $p_{\text{adj}} < 0.05$ on each failing encoder. Our stated prediction was that the anchor and gate carry most of the rescue.

Table 5 gives the result: **the rescue is the anchor, not the physics**. Anchor and gate alone account for 96–112% of the validation repair on each failing encoder. A plain residual head, with no gate and no simulator, already brings them to 17.47–17.84. Adding the SEIR simulator to the gated head earns credit on none of the three. It is significantly worse on STGAT (+0.67, 2/9, $p_{\text{adj}} = 0.047$), ties on A3TGCN (-0.00 , 6/9), and helps DCRNN by -0.68 on only 3/9 units ($p_{\text{adj}} = 0.38$). On the three working encoders the simulator costs 0.74–0.88 on 0/9 units each. The best head for AAGCN on

Table 3: Every lever, paired against its own control on matched (origin, seed) runs. Δ is validation RMSE (negative is better); wins count the 9 units where the arm is better; p_{adj} is BH across this table. Test Δ is shown and decided nothing. With three origins these p -values describe run-to-run stability, not the series (§4.4).

lever	Δ val	wins	p_{adj}	Δ test
<i>Worked</i>				
NB likelihood (vs squared error)	-0.41	9/9	0.018	+1.57
Seasonal features	-0.81	6/9	0.204	+1.07
Best model <i>B</i> vs persistence	-2.20	9/9	0.018	+1.54
<i>Graph and architecture</i>				
Graph (GCN vs no message passing)	-0.07	4/9	0.516	-0.20
Learned adjacency	-0.14	5/9	0.204	-0.26
District seasonal curves	-0.01	6/9	0.945	+0.48
Nonlinear (MLP) head	+0.26	2/9	0.049	+0.26
MLP head + climate lags 2–13	+0.01	3/9	0.945	+0.31
Global context node	-0.07	5/9	0.643	-0.36
Residual head + NB	+0.33	4/9	0.360	-2.99
Combined architecture changes	+0.32	2/9	0.321	+0.20
<i>Literature remedies</i>				
RevIN	+1.27	0/9	0.018	-0.81
District identity embedding	+0.06	4/9	0.744	+2.19
STID-style MLP	+0.71	2/9	0.055	-1.22
NB-GLM (no network)	+1.32	1/9	0.031	+5.30
k -NN analogues (no network)	+0.87	0/9	0.018	-3.20
Ensemble <i>B</i> + NB-GLM + k -NN	-0.01	5/9	0.945	-1.81
<i>Data</i>				
Half the training windows	+0.22	3/9	0.194	+1.07
TimeGAN augmentation	+0.53	1/9	0.055	-0.02
SEIR-simulated pre-training	-0.06	4/9	0.517	+0.27
Climate, lags 2–4	+0.09	3/9	0.516	+0.83
Climate, lags 2–13	+0.13	3/9	0.523	+0.05
Climate, lags 2–25	+0.30	2/9	0.055	-0.35
<i>Physics</i>				
SEIR as auxiliary loss	-0.04	6/9	0.516	-0.37
SEIR simulator vs no-physics twin (AAGCN)	+0.88	0/9	0.018	+2.03
Spatial physics penalty	-0.06	6/9	0.516	+0.50
Metapopulation vs gated SEIR head	+0.20	2/9	0.049	+0.20
Gated SEIR-GNN vs SEIR-LSTM	-0.17	9/9	0.018	-0.04

this grid is the gated head without physics (16.58 against the direct head’s 16.76; -0.18 , 8/9), a comparison we did not pre-register and report only as a lead. Our prediction was right that anchor and gate carry the repair (share > 0.8). It was wrong that the simulator’s effect would stay within ± 0.3 , and wrong that the residual head would rescue less than the gated one.

On test the picture reverses. The SEIR head beats its no-physics twin on the failing encoders (A3TGCN -1.65 and DCRNN -3.33 , both 9/9; STGAT -0.83 , 7/9). That is the recurring disagreement of §5.7. It was not used to decide anything, and we do not claim it.

5.3 What moves the metric

Table 3 pairs every lever we tried against its own control, and Figure 2 shows the same deltas. Two levers improve validation consistently. The **negative-binomial likelihood** is the largest controlled gain: -0.41 , 9/9, $p_{\text{adj}} = 0.018$, larger than any architecture

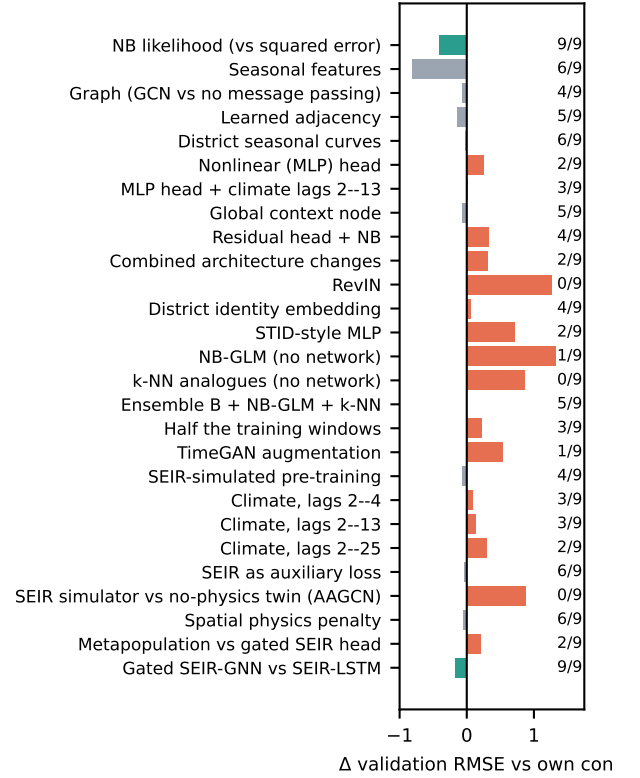


Figure 2: Every lever’s validation delta against its own control (Table 3); win counts on the right. Green: negative and $\geq 7/9$ wins. Grey: negative but inconsistent. Red: worse.

difference. It corrects a bias, since squared error on $\log(1+y)$ targets a median while RMSE scores a mean. **Seasonal features** give the largest mean change (-0.81) but win only 6/9. Combined, the best model *B* (AAGCN, direct head, NB, seasonal features) scores 15.66 against persistence’s 17.86: -2.20 on 9/9 units.

That is the headline a benchmark would report. It does not hold out of sample: on test, *B* scores 37.55 against persistence’s 36.02, and the NB likelihood is *worse* on test by 1.57. We return to this in §5.7.

5.4 SEIR-GNN against SEIR-LSTM

The comparison the research question poses is a graph encoder against the LSTM, behind the same gated SEIR head. On three origins the graph wins on validation every time: ASTGCN -0.62 (9/9), AAGCN -0.18 (9/9) with NB and seasonal features, and AAGCN -0.17 (9/9) in a separate grid. Behind the direct head the same graph gains nothing (-0.14 , 5/9). So what the graph adds appeared specific to estimating a transmission quantity, where spatial coupling is mechanistically plausible.

The nine-origin confirmation does not support that reading (Table 4). In the first run, the graph’s advantage holds on validation (-0.75 , 8/9, $p = 0.012$) and *reverses* on test ($+0.23$, 4/9). In the second, with a different encoder, it vanishes on validation (-0.01) and

Table 4: Nine disjoint origins, paired at the origin unit (the unit a claim about the series rests on; attainable p floor 0.004). Raw two-sided p . The two runs share the same test spans and are not independent replications.

comparison	run	validation			test		
		Δ	wins	p	Δ	wins	p
gated SEIR-GNN (ASTGCN) vs SEIR-LSTM	EXP-038	-0.75	8/9	0.012	+0.23	4/9	0.410
gated SEIR-GNN (AAGCN) vs SEIR-LSTM	EXP-047	-0.01	6/9	0.996	-0.41	6/9	0.031
metapopulation SEIR-GNN vs SEIR-LSTM	EXP-047	+0.51	4/9	0.344	-0.01	4/9	0.988
AAGCN vs LSTM, direct head	EXP-038	-0.95	7/9	0.531	-1.04	5/9	0.828
B -type direct model vs persistence	EXP-038	-2.89	8/9	0.035	+2.17	4/9	0.324
B vs persistence	EXP-047	-2.92	8/9	0.035	+1.45	4/9	0.367
gated SEIR-GNN vs persistence	EXP-047	-2.17	8/9	0.105	-0.70	6/9	0.652

Table 5: Anchor, gate or physics? Each encoder behind four heads that add, in turn, the persistence anchor, the learned gate and the SEIR simulator (mean validation RMSE; configuration of Table 2, all rerun together on Kaggle). The last two columns pair the gated SEIR head against its no-physics twin (gated); negative favours the physics, and the count is units where it wins.

encoder	direct	resid.	gated	SEIR	SEIR – gated	
					val	test
AAGCN	16.76	16.75	16.58	17.46	+0.88 (0/9)	+2.03 (0/9)
ASTGCN	16.83	16.78	16.74	17.48	+0.74 (0/9)	+0.95 (3/9)
LSTM	16.91	16.95	16.94	17.69	+0.74 (0/9)	-0.34 (6/9)
STGAT	23.32	17.84	16.95	17.62	+0.67 (2/9)	-0.83 (7/9)
A3TGCN	28.63	17.47	17.68	17.68	-0.00 (6/9)	-1.65 (9/9)
DCRNN	34.78	17.49	18.36	17.68	-0.68 (3/9)	-3.33 (9/9)
persistence		17.86				

appears on test (-0.41 , 6/9, $p = 0.031$). Neither run clears its pre-registered corrected threshold. The two confirmations also named different endpoints (validation for the first, test for the second), a flaw in our own pre-registration that we record rather than resolve after the fact. We therefore do not claim that the SEIR-GNN beats the SEIR-LSTM. On three origins it does so consistently on validation but not on test, where ASTGCN’s gated SEIR head is +0.30 against the SEIR-LSTM. On nine origins it is not robust to the encoder, the metric or the run.

5.5 Physics as structure: penalty and metapopulation coupling

Two further physics-informed structures were pre-registered against B . The **spatial physics penalty** was the one significant physics gain in our earlier work on the benchmark array (-0.044 on STGAT, 57/60 runs). On B it keeps its sign (-0.06 , and -0.20 at horizon 3) but wins only 6/9 and is not adopted. The **metapopulation SEIR head**, which couples districts inside the dynamics rather than through the initial state, is *worse* than the gated head ($+0.20$, 2/9). On nine origins it ties the SEIR-LSTM (-0.01 on test, 4/9). The reason is in the data. On the benchmark array, Moran’s I of log R on the district graph is 0.005, against 0.274 for log cases. Transmission has almost

no local spatial structure at district-week resolution, so a learned coupling has nothing to encode.

5.6 The ceiling is informational

Table 3 records what did not work. No architecture change improves validation by more than 0.14 (a learned adjacency, 5/9), and several make it worse: the nonlinear head by 0.26, the residual head with NB by 0.33 and all changes combined by 0.32. The literature’s remedies for weak spatio-temporal models are neutral or worse. RevIN costs 1.27 (0/9), and k -nearest-neighbour analogues with no network cost 0.87. Data-side levers fail too. Halving the training windows costs 0.22, and the learning curve is flat past 75%. TimeGAN augmentation costs 0.53 (1/9) and truncates the tail: its largest generated week is 882 cases against a real 2,631. Climate costs 0.09, 0.13 and 0.30 at lags 2–4, 2–13 and 2–25. Gradient-boosted trees gain from climate (-0.49), which a network does only once its head is nonlinear, and then the nonlinear head costs as much as climate gains.

These failures share a cause. The residuals of every model family we trained correlate 0.87–0.99 with B ’s, including k -NN, which has no network and no training, so the models make the same errors. The remaining error is a lag on moves. Models under-predict rising weeks by 11–13 cases and over-predict falling weeks by 10–11, and the top 5% of district-weeks carry ~60% of the squared error. That is what a mean-seeking objective produces on a near-random-walk target whose next move is not in the inputs.

5.7 Validation and test disagree, systematically

Across the twelve grids that share one window length, the arm validation selects is worse than persistence on test in eight, and the best arm on test is anchored to the last observation (gated SEIR, residual, gated or k -NN) in seven. Examples: the residual head with NB (34.55 against B ’s 37.55), k -NN (34.34), the gated SEIR arms (34.92–35.34), and on nine origins the gated SEIR-GNN at 31.06 against B at 33.21 and persistence at 31.76. Nothing here was selected on test, and none of these differences is significant on nine origins (the gated SEIR-GNN against B is -2.15 on test, 6/9, $p = 0.20$). A pattern across most grids is nonetheless evidence about the selection protocol. A 30-week validation span immediately before each origin appears to reward fitting that span’s level, which does not carry into the next period. Acting on it needs a new pre-registered protocol and a test set nothing here has touched, which only data after March 2024 can provide.

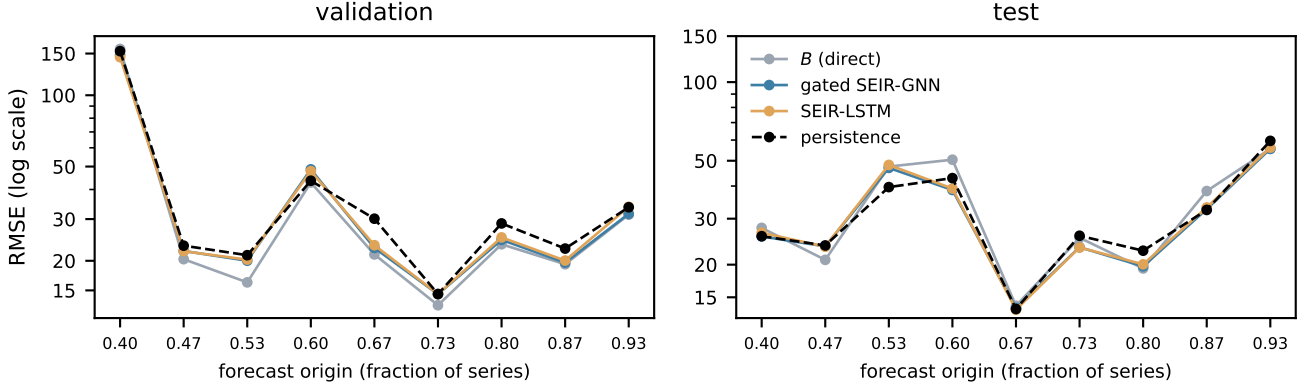


Figure 3: Nine disjoint origins (EXP-047), per-origin RMSE on a log scale. Origin 0.40’s validation span contains the 2017 DENV-2 epidemic, 5.4× anything in its training data, and dominates every validation mean; read win counts, not means. On test the gated SEIR arms track or undercut persistence while the direct model B is worst on four origins.

6 DISCUSSION

Where physics helps a graph network, and where it does not. On this target the SEIR model cannot be the forecaster. A pure SEIR decoder inherits the unpredictability of its force of infection, and 14–16% of targets lie out of its reach. The gated SEIR head seemed to show that physics could still help as structure, since it made every encoder usable and repaired three published architectures. The control in §5.2 removes that reading. The repair comes from anchoring the forecast to last week’s value, which a residual head does without any mechanism, and on validation the simulator adds nothing or costs up to 0.9 RMSE. Coupling districts inside the dynamics, as metapopulation GNNs do, does not help either, because transmission has almost no local spatial structure at district-week resolution. The prior that matters here is the one the data already state, that next week looks like this week.

Two findings keep the question open rather than closed. First, on test the SEIR head beats its no-physics twin on the failing encoders (up to -3.33 , 9/9), and every gated SEIR arm we trained sits at or below persistence on test, in every grid where it appears; the direct models do so only in some. Second, the graph’s advantage over the LSTM appears only inside the SEIR head. Neither survives our selection protocol, and we have argued that the protocol itself may be the reason (§5.7). A test of either needs data nothing here has touched.

Why the ceiling is not architectural. The target’s own last value explains $r^2 = 0.89$ of the next, the best covariate 0.03, and what remains is a move the inputs do not reveal. Every model family converges to the same function, with residual correlation 0.87–0.99. Architecture, embeddings, climate, synthetic data and physics penalties each improve validation by at most 0.14, and most make it worse. A study that searches this space and reports its best cell will find a winner; a paired study finds noise. The NB likelihood is the exception, because it corrects what the objective targets rather than what the model sees.

What the evaluation taught us. Each of the following would have produced a publishable positive result had it gone unchecked:

- the benchmark array leaks future data through its row order and its climate channels;
- our own first SEIR-GNN pipeline reported a win that compared one origin against a three-origin baseline and did not survive a correct training loop;
- the gated SEIR head’s repair of three encoders looked like physics and was the anchor;
- a validation span before each origin selects models that underperform anchored ones on test, in most of our twelve grids;
- two nine-origin confirmations disagree with each other.

Limitations. One country and one disease at district-week resolution. Three origins cannot support claims at the origin unit, and our nine-origin test spans were used twice, so they are not fresh. The two confirmations named different endpoints. Some earlier diagnostics (renewal predictability, Moran’s I , the STGAT spatial penalty) were measured on the benchmark array and have not been rerun on the corrected data. The graph is the benchmark’s hand-built adjacency; mobility data might couple districts differently. The SEIR head uses fixed stage durations and one initial susceptible fraction for every district.

Next steps. Clean confirmation needs data nothing here has touched: the weekly reports after March 2024. Before any such run we would pre-register a single endpoint, selection over several validation spans rather than one, and four finalists: B , the gated head without physics, the gated SEIR-GNN and the SEIR-LSTM. That design would settle whether the simulator’s test-set advantage is real.

7 CONCLUSION

We asked whether replacing the LSTM of an SEIR-informed forecaster with a graph network improves district-level dengue forecasts. Under a leak-free protocol on a corrected dataset, the answer is narrower than either literature suggests. The graph adds little on its own (0.07). The graph’s advantage over the LSTM inside the

SEIR head is consistent on three origins and not robust on nine. The largest controlled gain comes from the likelihood, not the architecture. Physics fails as a decoder for measurable, structural reasons, and its apparent success as a gated head belongs to the persistence anchor it is gated onto. Code, pre-registrations, the experiment log and a reproduction notebook accompany this paper.

REFERENCES

- [1] Jizhou Bai, Jiawei Zhu, Yanyan Song, Ling Zhao, Zhixiang Huang, and Ronghua Du. 2021. A3T-GCN: Attention Temporal Graph Convolutional Network for Traffic Forecasting. *ISPRS International Journal of Geo-Information* 10, 7 (2021), 485.
- [2] Qi Cao, Renhe Jiang, Chuang Yang, Zipei Fan, Xuan Song, and Ryosuke Shibasaki. 2023. Metapopulation Graph Neural Networks: Deep Metapopulation Epidemic Modeling with Human Mobility. *arXiv preprint arXiv:2306.14857* (2023).
- [3] Rasitha Banu GulMohamed, Wafa Hetany, Hanan Abdullah Almaimani, and Faiza Abdalla Saeed Khiery. 2026. DengueGNN: Graph-based deep learning for modeling disease spread dynamics and prediction. *Scientific Reports* 16 (2026), 10584. <https://doi.org/10.1038/s41598-026-43073-y>
- [4] Shengnan Guo, Youfang Lin, Ning Feng, Chao Song, and Huaiyu Wan. 2019. Attention Based Spatial-Temporal Graph Convolutional Networks for Traffic Flow Forecasting. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 33. 922–929.
- [5] Taesung Kim, Jinhee Kim, Yunwon Tae, Cheonbok Park, Jang-Ho Choi, and Jaegul Choo. 2022. Reversible Instance Normalization for Accurate Time-Series Forecasting against Distribution Shift. In *International Conference on Learning Representations (ICLR)*.
- [6] Yaguang Li, Rose Yu, Cyrus Shahabi, and Yan Liu. 2018. Diffusion Convolutional Recurrent Neural Network: Data-Driven Traffic Forecasting. In *International Conference on Learning Representations (ICLR)*.
- [7] Y. Liu, P. Fransson, J. Heidecke, P. Liyanage, J. Wallin, and J. Rocklöv. 2025. An explainable covariate compartmental model for predicting the spatio-temporal patterns of dengue in Sri Lanka. *PLoS Computational Biology* 21, 9 (2025), e1013540. <https://doi.org/10.1371/journal.pcbi.1013540>
- [8] Thang Nguyen, Dung Nguyen, Kha Pham, and Truyen Tran. 2024. MP-PINN: A Multi-Phase Physics-Informed Neural Network for Epidemic Forecasting. *arXiv preprint arXiv:2411.06781* (2024).
- [9] Ganga Ram Phaijoo and Dil Bahadur Gurung. 2018. Sensitivity Analysis of SEIR–SEI Model of Dengue Disease. *GAMS Journal of Mathematics and Mathematical Biosciences* 6, a (2018), 41–49.
- [10] Maziar Raissi, Paris Perdikaris, and George Em Karniadakis. 2019. Physics-Informed Neural Networks. *J. Comput. Phys.* 378 (2019), 686–707.
- [11] Alexander Rodríguez, Jiaming Cui, Naren Ramakrishnan, Bijaya Adhikari, and B. Aditya Prakash. 2023. EINNs: Epidemiologically-Informed Neural Networks. In *AAAI Conference on Artificial Intelligence*.
- [12] David Salinas, Valentin Flunkert, and Jan Gasthaus. 2020. DeepAR: Probabilistic Forecasting with Autoregressive Recurrent Networks. *International Journal of Forecasting* 36, 3 (2020), 1181–1191.
- [13] Zezhi Shao, Zhao Zhang, Fei Wang, Wei Wei, and Yongjun Xu. 2022. Spatial-Temporal Identity: A Simple yet Effective Baseline for Multivariate Time Series Forecasting. In *ACM International Conference on Information and Knowledge Management (CIKM)*.
- [14] Lei Shi, Yifan Zhang, Jian Cheng, and Hanqing Lu. 2019. Two-Stream Adaptive Graph Convolutional Networks for Skeleton-Based Action Recognition. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 12026–12035.
- [15] Hasitha A. Tissera, Bernard D. W. Jayamanne, Rajendra Raut, S. M. D. Janaki, Yesim Tozan, Preshila C. Samaraweera, Prasad Liyanage, Azhar Ghouse, Chaturaka Rodrigo, Aravinda M. de Silva, and Sumadhya D. Fernando. 2020. Severe Dengue Epidemic, Sri Lanka, 2017. *Emerging Infectious Diseases* 26, 4 (2020), 682–691.
- [16] Lijing Wang, Aniruddha Adiga, Jiangzhuo Chen, Adam Sadilek, Srinivasan Venkatramanan, and Madhav Marathe. 2022. CausalGNN: Causal-Based Graph Neural Networks for Spatio-Temporal Epidemic Forecasting. In *AAAI Conference on Artificial Intelligence*.
- [17] Jiaqi Weng, David Qiu, Ethan Cruz, Malik Magdon-Ismael, Thilanka Munasinghe, Jennifer C. Wei, and Ashan Pathirana. 2024. Graph Representation Learning for Dengue Forecasting. In *IEEE International Conference on Big Data (BigData)*.
- [18] Jinsung Yoon, Daniel Jarrett, and Mihaela van der Schaar. 2019. Time-series Generative Adversarial Networks. In *Advances in Neural Information Processing Systems (NeurIPS)*.